《数据库系统原理》实验报告（2）					
题目：DML 语言实验					
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实验环境：MariaDB

实验步骤及结果截图：

1.实验二中表结构和实验一相同，仅修改了 Student 表中关于 Gender 的约束，先修改约束

```
MariaDB [TeachingManagment]> alter table Student drop constraint GHK_Gender;
Query OK, 0 rows affected (0.022 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> ALTER TABLE Student MODIFY Gender VARCHAR(10) NOT NULL;
Query OK, 0 rows affected (0.010 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> ALTER TABLE Student
    -> ADD CONSTRAINT chk_gender_full
    -> CHECK (Gender IN ('Female', 'Male'));
Query OK, 0 rows affected (0.023 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> SELECT CONSTRAINT_NAME  FROM information_schema.CHECK_CONSTRAINTS  WHERE TABLE_NAME = 'Student';
+-----------------+
| CONSTRAINT_NAME |
+-----------------+
| chk_gender_full |
+-----------------+
1 row in set (0.005 sec)
```

2.插入数据

```
MariaDB [TeachingManagment]> INSERT INTO Dept VALUES
    ->     (1, 'Computer Science'),
    ->     (2, 'Mathematics'),
    ->     (3, 'English'),
    ->     (4, 'Management');
Query OK, 4 rows affected (0.007 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> INSERT INTO Student VALUES
    ->     (202415120, 'Michael', 'Male', 21, 3),
    ->     (202415121, 'Tommy', 'Male', 20, 1),
    ->     (202415122, 'Michael', 'Female', 19, 1),
    ->     (202415123, 'Ale', 'Female', 18, 2),
    ->     (202415125, 'Billy', 'Male', 19, 3);
Query OK, 5 rows affected (0.003 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> INSERT INTO Course VALUES
    ->     (1, 'Database', 5, 1),
    ->     (2, 'Mathematics', 2, 2),
    ->     (3, 'Information System', 1, 4),
    ->     (4, 'Operating System', 6, 1),
    ->     (5, 'Data Structure', 4, 1),
    ->     (6, 'Data Processing', 2, 4),
    ->     (7, 'PASCAL', 3, 1);
Query OK, 7 rows affected (0.003 sec)
Records: 7  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> INSERT INTO Score VALUES
    ->     (202415121, 1, 92),
    ->     (202415121, 2, 85),
    ->     (202415121, 3, 88),
    ->     (202415122, 2, 90),
    ->     (202415122, 3, 80);
```

3.查询所有年龄在 21 岁以下学生的姓名和年龄

```
MariaDB [TeachingManagment]> select Name,Age
    -> from Student
    -> where Age < 21;
+---------+-----+
| Name    | Age |
+---------+-----+
| Tommy   |  20 |
| Michael |  19 |
| Ale     |  18 |
| Billy   |  19 |
+---------+-----+
4 rows in set (0.008 sec)
```

4.查询选 2 号课程且成绩在 80－90 的学号

```
MariaDB [TeachingManagment]> select S_no
    -> from Score
    -> where C_no = 2 and Score between 80 and 90;
+-----------+
| S_no      |
+-----------+
| 202415121 |
| 202415122 |
+-----------+
2 rows in set (0.008 sec)
```

5.查询姓名第二个字母是 e 的学生姓名

```
MariaDB [TeachingManagment]> select Name
    -> from Student
    -> where Name like "_e%"
    -> ;
Empty set (0.018 sec)
```

6. 查询全体男学生的学号、系、年龄。结果按所在的系升序排列，同一系中的学生按年龄降序排列。

```
MariaDB [TeachingManagment]> SELECT No, D_no, Age
    -> FROM Student
    -> WHERE Gender = 'Male'
    -> ORDER BY D_no ASC, Age DESC;
+-----------+------+-----+
| No        | D_no | Age |
+-----------+------+-----+
| 202415121 |    1 |  20 |
| 202415120 |    3 |  21 |
| 202415125 |    3 |  19 |
+-----------+------+-----+
3 rows in set (0.005 sec)
```

7.查询女学生的总人数平均年龄

```
MariaDB [TeachingManagment]> select count(*) as num_female,avg(Age) as average_age from Student where Gender = "Female";
+------------+-------------+
| num_female | average_age |
+------------+-------------+
|          2 |     18.5000 |
+------------+-------------+
1 row in set (0.002 sec)
```

8. 查询选修 2 号课程并及格（分数大于 60）的学生最高分、最低分及总分。

```
MariaDB [TeachingManagment]> select max(Score) as max_score,min(Score) as min_score,sum(Score) as sum_score
    -> from Score
    -> where C_no = 2 and Score >60;
+-----------+-----------+-----------+
| max_score | min_score | sum_score |
+-----------+-----------+-----------+
|        90 |        85 |       175 |
+-----------+-----------+-----------+
1 row in set (0.006 sec)
```

9. 向 Score 表中插入一条记录 (200266666, 1, 85)。

```
MariaDB [TeachingManagment]> insert into Score
    -> values (200266666,1,85);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`TeachingManagment`.`Score`, CONSTRAINT `FK_Score_Student` FORE
IGN KEY (`S_no`) REFERENCES `Student` (`No`))
```

10. 求每个学生（号）的平均成绩，并将其中超过 70 分的按学号输出

```
MariaDB [TeachingManagment]> select S_no,avg(Score) as average_score
    -> from Score
    -> group by S_no
    -> having average_score > 70
    -> order by S_no;
+-----------+---------------+
| S_no      | average_score |
+-----------+---------------+
| 202415121 |       88.3333 |
| 202415122 |       85.0000 |
+-----------+---------------+
2 rows in set (0.006 sec)
```

11. 查询选修了课程 1 或者选修了课程 3 的学生姓名

```
MariaDB [TeachingManagment]> select distinct S.Name from Student as S join Score as Sc on S.No = Sc.S_No where Sc.C_no in (1,3);
+---------+
| Name    |
+---------+
| Tommy   |
| Michael |
+---------+
2 rows in set (0.010 sec)
```

12. 查询既选修了课程 1 又选修了课程 2 的学生姓名

```
MariaDB [TeachingManagment]> select name
    -> from Student
    -> where No in(
    -> select S1.S_no
    -> from Score as S1
    -> join Score as S2
    -> where S1.C_no = 1 and S2.C_no = 2);
+-------+
| name  |
+-------+
| Tommy |
+-------+
1 row in set (0.009 sec)
```

13. 查询选修 Database 这门课最高分学生所在的系名

```
MariaDB [TeachingManagment]> SELECT D.Name FROM Dept D JOIN Student S ON D.No = S.D_no JOIN Score SC ON S.No = SC.S_no JOIN Course C ON SC.C_no = C.N
o WHERE C.Name = 'Database' ORDER BY SC.Score DESC LIMIT 1;
+-----------------+
| Name            |
+-----------------+
| Computer Science |
+-----------------+
1 row in set (0.011 sec)
```

14. 建立一个包含学生学号，姓名，年龄，以及所在系名的视图（赋予列名为 sno, sname, sage, deptname）

```
MariaDB [TeachingManagment]> CREATE VIEW StudentInfo (sno, sname, sage, deptname) AS
    ->
    -> SELECT S.No, S.Name, S.Age, D.Name
    ->
    -> FROM Student S
    ->
    -> JOIN Dept D ON S.D_no = D.No;
Query OK, 0 rows affected (0.023 sec)
```

出现的问题：

解决查询选修 Database 这门课最高分学生所在的系名时想使用函数 max 来实现，但没有成功

解决方案：

最后使用排序加 limit 关键字，找到最高分的行